



UNIVERSITY OF SCIENCE - VNUHCM

FACULTY OF INFORMATION TECHNOLOGY

SOFTWARE TESTING

HW5 - AUTOMATION TESTING

Software Testing Project Report

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1 Group Information

Group ID: 07

Member Name	Student ID	Assigned Features	Status
Cao Uyển Nhi	22127310	- SignUp	Done
		- Checkout	Done
Lưu Thanh Thuý	22127410	- SignIn	Done
		- User Management	Done
Nguyễn Phước Minh Trí	22127424	- Catalog	Done
		- Categories	Done
Võ Lê Việt Tú	22127435	- MyProfile	Done
		- Order Management	Done
Trần Thị Cát Tường	22127444	- Contact	Done
		- Category Management	Done

2 Automation Testing Summary

2.1 Test Data Preparation

2.1.1 Contact:

- Test cases stored in 22127444_Data.xlsx, sheet: **Contact**.
- Key fields: First Name, Last Name, Email, Subject, Message, Attachment, Is Login, Expected Result.
- Coverage includes:
 - Valid submission (both guest and logged-in user)
 - Missing required fields
 - Edge cases: invalid email, long input, oversized files, wrong file format

2.1.2 Category Management:

- Test cases in the same Excel file, sheet: **AddCategory**.
- Fields: Name, Slug, Parent ID, Expected Result.
- Covers:
 - Duplicate name/slug
 - Missing name/slug
 - Invalid characters in slug
 - Max length enforcement
 - Parent-child relationship

2.2 Script Initialization

- Environment setup:

```
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
```
- WebDriver setup for Chrome, Firefox, and Edge.
- Logs and screenshots are saved under logs/ directory.
- Data loaded via `pandas.read_excel()`.

2.2.1 Test Execution Logic

Contact Tests:

- Auto-login using `customer@practicesoftwaretesting.com` if `Is Login = TRUE`.
- Navigate to `/contact` after login (or directly if guest).
- If guest:
 - Fill in: First Name, Last Name, Email.

- Select Subject from dropdown if matched.
- Fill Message field if applicable.
- Attach file if it exists locally.
- Click Submit and validate:
 - If successful, check if “Thanks for your message” is included in the response body.
 - Otherwise, validate each line in `expected_result` is present in the page body.
 - Use `normalize()` to handle case and character differences.

Category Tests:

- Auto-login as Admin (`admin@practicesoftwaretesting.com`).
- Navigate to `/admin/categories/add`.
- Fill in: Name, Slug.
- If Parent ID is provided:
 - Check if value exists in dropdown before selecting.
- Submit and validate:
 - If `expected_result` includes “category added”, verify body contains expected success message.
 - Else, validate all expected texts appear in the response.
 - Use normalization logic similar to contact form.

2.3 Mapping: Excel to UI Fields

Excel Column	UI Field
First_name	ID: first_name
Subject	Dropdown: subject
Attachment	File input: attachment
Name (category)	ID: name
Slug	ID: slug
Parent ID	Select: parent_id

2.4 Checkpoint Logic

2.4.1 Contact:

Expected Result	Checkpoint Condition
Thanks for your message! We will contact you shortly.	Body text includes success message
Invalid email	Page still present, error message for email displayed

Expected Result	Checkpoint Condition
The email field must not be greater than 120 characters	Page still present, error message includes max length info
The name field must not be greater than 120 characters	Error message appears under the corresponding field
Message too long	Error message shown for message field
File too large or wrong format	Error for attachment input shown (e.g. “invalid file type”)

2.4.2 Category:

Expected Result	Checkpoint Condition
Category is added successfully	Body text or alert includes “category saved” or similar
Name is required	Error shown near name input field
Name must be at most 128 characters	Validation message below name input
Name must be unique	UI shows “category name already exists”
Slug must be URL-safe	Error message: slug must be lowercase/hyphen-only/no spaces
Slug already exists	UI shows “slug already exists”
Slug must be at most 128 characters	Validation error below slug field
Parent category assigned correctly	New category appears under selected parent (if applicable)

2.5 Result Overview

2.5.1 Contact Tests:

- **Total:** 30
- **PASS:** 11
- **FAIL:** 19 (mainly due to message mismatch or missing file validations)

2.5.2 Category Tests:

- **Total:** 23
- **PASS:** 2
- **FAIL:** 21 (majority due to UI messages not matching expected string exactly)

2.6 Notes & Suggestions

- Consider normalizing messages from UI using `.lower()` and trimming spaces to reduce brittle string comparison.
- Investigate mismatch between expected result and actual success message (e.g., ‘category saved’ vs ‘category is added successfully’).

- Improve test data consistency: unify all success messages and clarify edge case boundaries.
- Validate subject options dynamically for contact form to avoid test failure due to UI changes.

2.7 Video Demo

- **Contact Feature:** [Watch here](#)
- **Category Management:** [Watch here](#)

3 Self-Evaluation

No.	Criteria	Grade	Self-Assessed
1	Contact Feature	50	50
	Report + Bug Analysis	15	15
	Test Cases	5	5
	Script Files	10	10
	Data Files	10	10
	Video	10	10
2	Category Management	50	50
	Report + Bug Analysis	15	15
	Test Cases	5	5
	Script Files	10	10
	Data Files	10	10
	Video	10	10
	Total	100	100